

UNIVERSITY OF CAPE COAST
COLLEGE OF DISTANCE EDUCATION (CoDE)
DEPARTMENT OF SCIENCE AND MATHEMATICS
INF811D: OBJECT ORIENTED PROGRAMMING
MSc Information Technology Programme
SEMESTER 1 PAPER / PRACTICAL PROJECT
2025/2026 ACADEMIC YEAR

INSTRUCTIONS

1. This semester paper/project is to be completed individually.
2. Each student is required to select ONLY ONE project topic from the list provided below.
3. Students are required to design and implement a fully functional **Graphical User Interface (GUI)-Based Java Application**.
4. The project MUST be developed using:
 - a) JavaFX, or
 - b) Java Swing
5. Console-based applications are NOT permitted for this project.
6. The application must demonstrate a strong understanding and practical application of:
 - a) Object-Oriented Programming principles
 - b) Operators and expressions
 - c) Control structures
 - d) Methods and modular programming
 - e) Exception handling
 - f) Arrays and/or collections
 - g) Event-driven programming
 - h) GUI component handling
 - i) Input validation

7. The system must clearly demonstrate the implementation of the following Object-Oriented Programming concepts:
 - a) Encapsulation
 - b) Inheritance
 - c) Polymorphism
 - d) Abstraction
8. Students are encouraged to apply creativity, usability, and innovation in GUI design.
9. The project interface should be:
 - a) User-friendly
 - b) Well-organized
 - c) Professionally designed
 - d) Easy to navigate
10. Students must ensure that:
 - Programs compile and execute correctly
 - GUI controls function properly
 - Exception handling is implemented appropriately
 - Code is properly structured and documented
 - Meaningful variable, method, and class names are used
11. The completed project **MUST** be deployed to GitHub.
12. Students are required to:
 - Create a GitHub repository for the project
 - Push all source code to GitHub
 - Include a well-written README.md file
 - Include screenshots of the application in the repository
 - Maintain proper commit history throughout development
13. Students must submit:
 - GitHub repository link

- Technical report (PDF or Word format)
 - Screenshots of the application
 - Explanation of OOP concepts implemented
14. Students are expected to produce original work and write-ups. Reports should reflect individual understanding, analysis, and implementation experience. Students are therefore advised to avoid generic AI-generated content and ensure that explanations, discussions, screenshots, reflections, and documentation genuinely represent their own work and project development process.
15. Any form of plagiarism, repository sharing, copying or academic dishonesty will attract severe academic penalties.

TECHNICAL REPORT FORMAT

Students are required to prepare a technical report containing the following sections:

1. Title Page
2. Introduction
3. Problem Statement
4. Objectives of the System
5. Scope of the System
6. Methodology
7. System Design
8. Description of Classes and Methods
9. GUI Design Explanation
10. OOP Concepts Implemented
11. Screenshots and Outputs
12. GitHub Repository Link
13. Challenges Encountered
14. Conclusion
15. Recommendations
16. References

PROJECT REQUIREMENTS

Each project must satisfy the following minimum requirements:

Requirement	Marks
Functionality of System	20
GUI Design and User Experience	15
Application of OOP Concepts	20
Operators and Control Structures	10
Methods and Modularization	10
Event Handling and GUI Interaction	10
Exception Handling and Validation	5
GitHub Deployment and Documentation	5
Technical Report	5
TOTAL	100

PROJECT TOPICS

Education and University Systems

1. Smart Student Record Management System
2. Course Registration and Timetable System
3. Hostel Allocation Management System
4. Student Attendance Monitoring System
5. Online Examination and Grading System
6. Lecturer Evaluation Management System
7. School Fees Payment Tracking System
8. Research Project Supervision System
9. Student Internship Placement System
10. Academic Transcript Generation System

Healthcare Systems

11. Intelligent Hospital Queue Management System
12. Patient Medical Record System
13. Pharmacy Inventory Management System
14. Hospital Appointment Booking System
15. Blood Donation Management System
16. Vaccination Tracking System
17. Ambulance Dispatch Management System
18. Medical Laboratory Test Processing System
19. Clinic Billing and Payment System
20. Maternal Health Monitoring System

Banking and Finance Systems

21. ATM and Banking Simulation System
22. Mobile Money Transaction System
23. Payroll and Tax Processing System
24. Loan Management System
25. Savings and Investment Tracking System
26. Cooperative Credit Union System
27. Insurance Claims Management System
28. Cryptocurrency Portfolio Tracker
29. Expense Tracking and Budgeting System
30. Point of Sale (POS) System

Security and Government Systems

31. Immigration Border Entry Management System
32. National ID Verification System
33. Crime Reporting and Tracking System
34. Prison Inmate Management System

35. Police Case Management System
36. Passport Application Processing System
37. Fire Service Incident Reporting System
38. Smart Traffic Offence Management System
39. Vehicle Registration and Licensing System
40. Cybersecurity Incident Reporting System

Business and Enterprise Systems

41. Hotel Reservation Management System
42. Restaurant Ordering and Billing System
43. Supermarket Inventory Management System
44. Employee Leave Management System
45. Human Resource Management System
46. Warehouse and Stock Control System
47. Customer Relationship Management (CRM) System
48. Online Food Delivery Management System
49. Event Ticket Booking System
50. Delivery and Logistics Tracking System

GITHUB REQUIREMENTS

Students must ensure that their GitHub repository contains:

- Complete source code
- Proper folder structure
- README.md documentation
- Screenshots of the system
- Instructions on how to run the application
- Meaningful commit messages
- Appropriate naming conventions

IMPORTANT NOTES

- Students are strongly advised to start early.
- Projects should reflect MSc-level critical thinking and software engineering practices.
- Poor GUI design and weak OOP implementation will attract low marks.
- Students are encouraged to implement additional intelligent features where possible.
- Students are encouraged to write concise, technical, and original reports that accurately describe their own implementation process and experiences.
- Late submissions will not be encouraged.

SUBMISSION DETAILS

- Submission Deadline: **24/05/2026**
- Mode of Submission: GitHub Repository Link and Technical Report
- Technical Report Format: PDF or Microsoft Word
- Demonstration/Presentation Date: To Be Announced